A Corrected Proof of Respecting Improvement for Best Core Allotments in Unbounded Shapley–Scarf Housing Markets

Clean verification note

June 24, 2026

Abstract

This note repairs a proposed proof of the respecting-improvement property for the core of an unbounded Shapley–Scarf housing market with weak preferences. The repaired argument proves a complete solution, not merely partial progress: for every old core allocation, after an improvement of agent p's endowed object, there is a new core allocation in which p receives a weakly better object. Consequently p's best attainable core allotment cannot become worse. The result is not new; it is the weak-order specialization of the stronger theorem of Schlotter, Biró, and Fleiner. The purpose of the note is to give a self-contained, formally consistent proof in the classical complete-preference notation and to make explicit the dummy-vertex and final-acyclicity arguments that were missing from the draft under review.

1 Status of the result

The result below is a full solution of the RI-best question for the core when trading cycles are unbounded. It should not be presented as original: Schlotter, Biró, and Fleiner prove the stronger allocation-by-allocation statement for housing markets with partially ordered preferences, and derive the RI-best corollary. The proof here follows the same HM-Improve idea but is written in the complete weak-order model.

2 Housing markets and improvements

Let N be a finite set of agents. Agent i's endowed object is also denoted by i. A housing market is a pair

$$H = (N, (\succsim_i)_{i \in N}),$$

where each $\succsim_i$ is a complete transitive weak order on N. We write $x \succ_i y$ for the strict part of $\succsim_i$. Thus $x \succ_i y$ means that agent i strictly prefers object x to object y.

An allocation is a permutation X of N; $X(i) = j$ means that agent i receives object j. A nonempty coalition $S \subseteq N$ blocks X if there is a permutation π of S such that

$$\pi(i) \succ_i X(i) \qquad \text{for every } i \in S.$$

The core $\mathcal{C}(H)$ is the set of allocations that are not blocked.

For an allocation X, define its envy digraph G_X^H on vertex set N by

$$(i, j) \in E(G_X^H) \quad \Longleftrightarrow \quad j \succ_i X(i).$$

Loops are allowed. A loop (i, i) says that i strictly prefers her own object to the object she receives.

Lemma 1 (Core as acyclicity)**.** *An allocation X belongs to $\mathcal{C}(H)$ if and only if G_X^H has no directed cycle, loops included.*

Proof. If G_X^H has a directed cycle $i_1, i_2, \ldots, i_k, i_1$, then the coalition $\{i_1, \ldots, i_k\}$ blocks X by giving agent i_t object i_{t+1}, with indices modulo k. This also covers $k = 1$.

Conversely, suppose that S blocks X through a permutation π of S. Decompose π into disjoint directed cycles. On each cycle $i \mapsto \pi(i)$, every agent i strictly prefers $\pi(i)$ to $X(i)$. Hence that cycle is present in G_X^H. Thus blocking coalitions are exactly directed envy cycles. □

We use the following formal version of an improvement. Fix $p \in N$. A market $H' = (N, (\succsim_i')_{i \in N})$ is a *p-improvement* of H if $\succsim_p' = \succsim_p$, and for every $i \neq p$:

(I1) the relative order of all objects different from p is unchanged, i.e. for all $x, y \in N \setminus \{p\}$,

$$x \succsim_i y \iff x \succsim_i' y;$$

(I2) object p does not move downward: for every $x \in N \setminus \{p\}$,

$$x \succ_i' p \Rightarrow x \succ_i p, \qquad p \succ_i x \Rightarrow p \succ_i' x.$$

Equivalently, the only strict comparisons that can be newly created are comparisons in which p's object is strictly preferred to some other object.

For any nonempty set $\mathcal{A}$ of allocations, define the set of p-best objects attainable in $\mathcal{A}$ by

$$B_p(\mathcal{A}) := \{X(p) : X \in \mathcal{A} \text{ and } X(p) \succsim_p Y(p) \text{ for every } Y \in \mathcal{A}\}.$$

This set is nonempty because N is finite.

We shall use only the following standard existence fact.

Lemma 2 (Core existence)**.** *Every finite housing market with weak preferences has a core allocation. The same is true for every induced submarket.*

Proof. Run top trading cycles with arbitrary tie-breaking among currently most preferred remaining objects. At each round, every remaining agent points to one most preferred remaining object; a directed cycle exists and all agents on chosen cycles are assigned along those cycles and removed. The process terminates with an allocation.

If a coalition blocked the resulting allocation, choose an agent i in the coalition who was removed earliest by the algorithm. At that round all objects owned by the blocking coalition were still present. Agent i received one of her most preferred remaining objects, so she cannot strictly prefer the coalition object assigned to her in the alleged block. This contradiction proves that the allocation is in the core. The same argument applies after restricting the market to any subset of agents. □

3 The strong improvement theorem

The next theorem is the substantive statement. It is stronger than RI-best because it starts from an arbitrary old core allocation, not necessarily one that is best for p.

Theorem 3 (Strong improvement theorem)**.** *Let H' be a p-improvement of H, and let $X \in \mathcal{C}(H)$. Then there exists an allocation $X' \in \mathcal{C}(H')$ such that*

$$X'(p) \succsim_p X(p).$$

3.1 Suballocations

The proof uses a temporary directed graph with dummy vertices. A *suballocation* in a digraph $D = (M, E)$, from a set of sources $U \subseteq M$ to a set of sinks $V \subseteq M$ with $|U| = |V|$, is a set $Y \subseteq E$ satisfying

$$\begin{aligned} \delta^+_Y(v) &= 0 \quad \text{for } v \in V, \\ \delta^+_Y(a) &= 1 \quad \text{for } a \in M \setminus V, \\ \delta^-_Y(u) &= 0 \quad \text{for } u \in U, \\ \delta^-_Y(a) &= 1 \quad \text{for } a \in M \setminus U. \end{aligned}$$

Thus Y is a disjoint union of directed cycles and directed paths from U to V. If $a \notin V$, then $Y(a)$ denotes the unique head of the arc leaving a.

Given auxiliary preferences on the tails of arcs, an arc $(a, b) \in E$ is *Y-augmenting* if either $a \in V$, or $a \notin V$ and $b \succ^*_a Y(a)$. The envy graph of Y is the subgraph of D consisting of all Y-augmenting arcs. The suballocation Y is *stable* if this envy graph is acyclic.

3.2 Auxiliary digraph and algorithm

If $X \in \mathcal{C}(H')$, then Theorem 3 is immediate with $X' = X$. Assume from now on that $X \notin \mathcal{C}(H')$. By Lemma 1, $G^{H'}_X$ has a directed cycle. Since G^H_X is acyclic and a p-improvement can only create new strict envy arcs pointing to p, the set

$$Q := \{q \in N \setminus \{p\} : p \succ'_q X(q) \text{ and not } p \succ_q X(q)\}$$

is nonempty.

For every $q \in Q$, add a dummy vertex $\widehat{q}$. Let

$$\widehat{Q} := \{\widehat{q} : q \in Q\}, \qquad \widehat{N} := N \cup \widehat{Q}.$$

We now define an auxiliary digraph $\widehat{D} = (\widehat{N}, \widehat{E})$. For original vertices not in Q, all arcs to original objects are present:

$$(i, j) \in \widehat{E} \quad (i \in N \setminus Q, j \in N).$$

For $q \in Q$, all arcs from q to original objects other than p are present, the arc (q, p) is omitted, and the dummy arc $(q, \widehat{q})$ is added:

$$(q, j) \in \widehat{E} \quad (j \in N \setminus \{p\}), \qquad (q, \widehat{q}) \in \widehat{E}, \qquad (q, p) \notin \widehat{E}.$$

Finally, each dummy has exactly one outgoing arc:

$$(\widehat{q}, p) \in \widehat{E} \qquad (q \in Q),$$

and there are no other arcs incident with dummy vertices except the arcs just listed. Thus the only arc entering $\widehat{q}$ is $(q, \widehat{q})$, and the only arc leaving $\widehat{q}$ is $(\widehat{q}, p)$.

The auxiliary preferences are as follows. If $i \in N \setminus Q$, then $\succsim^*_i$ is $\succsim'_i$ on N. If $q \in Q$, then $\widehat{q}$ takes exactly the position that p has in q's improved preference order: with

$$\phi_q(\widehat{q}) = p, \qquad \phi_q(x) = x \quad (x \in N \setminus \{p\}),$$

we define, for $x, y \in (N \setminus \{p\}) \cup \{\widehat{q}\}$,

$$x \succsim^*_q y \quad \Longleftrightarrow \quad \phi_q(x) \succsim'_q \phi_q(y).$$

For a dummy $\widehat{q}$, no comparison is needed except that after it receives p, the arc $(\widehat{q}, p)$ is not strictly improving.

Initialize

$$Y_0 := X \setminus \{(q, X(q)) : q \in Q\} \ \cup \ \{(q, \widehat{q}) : q \in Q\}.$$

This is a suballocation in $\widehat{D}$ with source set

$$U_0 = \{X(q) : q \in Q\}$$

and sink set

$$V_0 = \widehat{Q}.$$

Let the irrelevant set initially be $R_0 = \varnothing$. The algorithm repeatedly modifies (Y, U, V, R) inside the current graph $\widehat{D} - R$. Initially $(Y, U, V, R) = (Y_0, U_0, V_0, R_0)$.

While $U \neq V$, do the following.

Step 1. If there is a Y-augmenting arc (s, u) entering some source $u \in U$, choose one such arc.

(a) If $s \notin V$, let $u' = Y(s)$. Replace (s, u') by (s, u) in Y, and set

$$U \leftarrow (U \setminus \{u\}) \cup \{u'\}, \qquad V \leftarrow V.$$

(b) If $s \in V$, add (s, u) to Y, and set

$$U \leftarrow U \setminus \{u\}, \qquad V \leftarrow V \setminus \{s\}.$$

Step 2. If no Y-augmenting arc enters any source, choose any $u \in U \setminus V$, let $u' = Y(u)$, delete (u, u') from Y, put u into R, and set

$$U \leftarrow (U \setminus \{u\}) \cup \{u'\}, \qquad V \leftarrow V.$$

The degree conditions show after each operation that Y is again a suballocation from the updated U to the updated V in $\widehat{D} - R$. Notice also that $V \subseteq \widehat{Q}$ throughout the algorithm.

Lemma 4 (Initial and maintained stability)**.** *At every stage of the algorithm, Y is stable in $\widehat{D} - R$.*

Proof. First consider the initial suballocation Y_0. No directed cycle in its envy graph can contain a dummy $\widehat{q}$, because the only arc entering $\widehat{q}$ is $(q, \widehat{q})$, and this arc is an arc of Y_0, hence is not Y_0-augmenting.

Thus any initial envy cycle would use only original vertices. Let (i, j) be an arc of such a cycle. If $i \notin Q$, then $Y_0(i) = X(i)$ and $j \succ'_i X(i)$. If $j \neq p$, the same strict comparison holds in H; if $j = p$, then the comparison is either old already or else i would belong to Q. Hence (i, j) is an old envy arc. If $i = q \in Q$, then $j \neq p$, since $(q, p) \notin \widehat{E}$, and $j \succ^*_q \widehat{q}$. By the definition of $\succ^*_q$, this means $j \succ'_q p$. Since $p \succ'_q X(q)$ by the definition of Q, transitivity gives $j \succ'_q X(q)$. Both j and $X(q)$ are different from p, so their relative order is unchanged by the improvement, and therefore $j \succ_q X(q)$. Again (q, j) is an old envy arc. Hence any initial envy cycle would be a directed cycle in G^H_X, contradicting $X \in \mathcal{C}(H)$ and Lemma 1.

Now assume stability before an iteration. In Step 1(a), the tail s receives u, which it strictly prefers to its old assignment $u' = Y(s)$. Therefore the set of outgoing augmenting arcs from s can only shrink; all other tails have the same assigned object, and the sink set is unchanged. In Step 1(b), s ceases to be a sink and receives u; before the step, every outgoing arc of s was augmenting, and after the step only a subset can be augmenting. In Step 2, the vertex u is deleted into R. Thus each operation obtains the new envy graph from the old one by deleting arcs or deleting a vertex. Acyclicity is preserved. ∎

Lemma 5 (Termination)**.** *The algorithm terminates.*

Proof. By the proof of Lemma 4, every nonterminal iteration deletes at least one arc from the current envy graph or deletes one vertex into R, and no operation creates a new envy arc. Since $\widehat{D}$ is finite, only finitely many such deletions can occur. $\square$

Lemma 6 (Monotonicity before irrelevance)**.** *Let $a \in N$ be an original vertex that is not in R at the end of the algorithm. If a has assigned objects at two stages of the algorithm, then its later assigned object is weakly preferred by a, in the auxiliary preference $\succsim_a^*$, to its earlier assigned object.*

Proof. An original vertex is never a sink, because $V \subseteq \widehat{Q}$ throughout. Hence, unless it is put into R, it always has an outgoing assignment. The only operation that changes the assigned object of an already assigned tail is Step 1(a), and there the new object is chosen by a strictly improving arc. The claim follows by induction over the iterations. $\square$

Lemma 7 (No final entry into the irrelevant set)**.** *At termination, there is no arc $(a, b) \in \widehat{E}$ with $a \in N \setminus R$, $b \in R$, and*

$$b \succ_a^* Y(a).$$

Proof. Suppose that such an arc exists at termination. Let b have been added to R in some execution of Step 2, and let Y_t be the suballocation at the beginning of that execution. At that moment b was a source, and no augmenting arc entered any source.

The original vertex a is not in R at the end, hence it was not in R at time t. By Lemma 6, its final assignment $Y(a)$ is weakly preferred to its assignment $Y_t(a)$ at time t. Since $b \succ_a^* Y(a)$, transitivity gives $b \succ_a^* Y_t(a)$. Therefore (a, b) was a Y_t-augmenting arc entering the source b, contradicting the condition for Step 2. $\square$

Lemma 8 (Agent p is never irrelevant)**.** *At termination, $p \notin R$.*

Proof. Suppose that p is put into R in Step 2. Just before that step, $p \in U \setminus V$. Since $|U| = |V|$, $V \subseteq \widehat{Q}$, and $U \neq V$, there is a dummy $\widehat{q} \in V \setminus U$. The dummy $\widehat{q}$ is a sink, and its unique outgoing arc is $(\widehat{q}, p)$. Hence $(\widehat{q}, p)$ is a Y-augmenting arc entering the source p, contradicting the assumption needed to execute Step 2. $\square$

3.3 Constructing the final allocation

Let Y, U, V, R be the objects at termination. Then $U = V \subseteq \widehat{Q}$. Let $R_N := R \cap N$ and $S := N \setminus R_N$. By Lemma 8, $p \in S$.

Because $U = V$, the suballocation Y consists of directed cycles covering all vertices of $\widehat{N} \setminus (R \cup U)$, while the vertices of U have no incident arcs of Y. Delete all dummy vertices. If no dummy lies on a cycle of Y, this leaves an allocation Z of the original agents in S. If a dummy $\widehat{q}$ lies on a cycle of Y, then its unique incoming and outgoing arcs are necessarily

$$(q, \widehat{q}), \qquad (\widehat{q}, p).$$

There is at most one such dummy, because Y has indegree one at p. In this case replace the two arcs above by (q, p), and then delete all dummies. The result is again an allocation Z of the original agents in S.

By Lemma 2, choose an allocation $X_R \in \mathcal{C}(H'[R_N])$ for the submarket induced by R_N, with $X_R = \varnothing$ if $R_N = \varnothing$. Define

$$X' := Z \cup X_R.$$

This is a permutation of N.

We now prove that $X' \in \mathcal{C}(H')$. The following claim is the key missing point in the informal draft.

Lemma 9 (Original envy arcs outside R)**.** *Let $a, b \in S$. If (a,b) is an envy arc of X' in H', then (a,b) is a final Y-augmenting arc of $\widehat{D} - R$.*

Proof. Assume $b \succ'_a X'(a)$. First suppose that it is not the case that $a \in Q$ and $b = p$. Then $(a,b) \in \widehat{E}$. If $Y(a) \in N$, then $Y(a) = X'(a)$, and the arc is Y-augmenting immediately. The only remaining possibility is $a = q \in Q$ and $Y(q) = \widehat{q}$. Then the dummy $\widehat{q}$ is the internal dummy that was spliced out, so $X'(q) = p$. Since $b \neq p$ in the present case and $b \succ'_q p$, the definition of $\succ^*_q$ gives $b \succ^*_q \widehat{q} = Y(q)$. Thus (q,b) is Y-augmenting.

It remains to handle the omitted arc (q,p) with $q \in Q$. We prove that this arc is never an envy arc of X'. If $Y(q) = \widehat{q}$ at termination, then the dummy is spliced and $X'(q) = p$, so $p \not\succ'_q X'(q)$. If $Y(q) \neq \widehat{q}$, then the initial arc $(q, \widehat{q})$ must have been deleted at some earlier step. Since $q \notin R$, this deletion could not have occurred by putting q into R in Step 2; it occurred in Step 1(a), when q replaced $\widehat{q}$ by some object u with $u \succ^*_q \widehat{q}$. By Lemma 6, the final assignment $Y(q) = X'(q)$ is weakly preferred to u in $\succsim^*_q$. Hence $Y(q) \succ^*_q \widehat{q}$. Translating $\widehat{q}$ back to p, this says $X'(q) \succ'_q p$, so again (q,p) is not an envy arc of X'. ☐

Lemma 10 (The output is core-stable)**.** *The allocation X' belongs to $\mathcal{C}(H')$.*

Proof. By Lemma 1, it is enough to show that the envy graph of X' in H' is acyclic.

The subgraph induced by R_N is the envy graph of X_R in the submarket $H'[R_N]$, and is acyclic because $X_R \in \mathcal{C}(H'[R_N])$. The subgraph induced by S is acyclic by Lemma 9 and Lemma 4: any directed envy cycle inside S would be a directed cycle in the final envy graph of Y in $\widehat{D} - R$.

It remains to rule out a cycle using vertices from both S and R_N. Such a cycle would contain an envy arc from some $a \in S$ to some $b \in R_N$. Since $p \notin R_N$, we have $b \neq p$. If $Y(a) \in N$, then $Y(a) = X'(a)$, and the envy relation $b \succ'_a X'(a)$ gives $b \succ^*_a Y(a)$. If $a = q \in Q$ and $Y(q) = \widehat{q}$, then $X'(q) = p$, and $b \neq p$ together with $b \succ'_q p$ gives $b \succ^*_q \widehat{q} = Y(q)$. In either case (a,b) is an auxiliary augmenting arc from an original vertex outside R into R, contradicting Lemma 7. Therefore no mixed directed envy cycle exists. ☐

Proof of Theorem 3. The construction above gives an allocation $X' \in \mathcal{C}(H')$ by Lemma 10. It remains only to compare p's object.

By Lemma 8, p is never put into R. Since p is an original vertex, it is never a sink, so it has an assigned object throughout the algorithm. Initially, $Y_0(p) = X(p)$. At the end, the assignment of p is unchanged by deleting or splicing dummy vertices, so $Y(p) = X'(p)$. By Lemma 6, $X'(p) \succsim^*_p X(p)$. Because p's own preference did not change and no dummy can be assigned to p, this is exactly $X'(p) \succsim_p X(p)$. ☐

4 RI-best follows

Corollary 11 (Respecting improvement for best core allotments)**.** *Let H' be a p-improvement of H. For every*

$$b \in B_p(\mathcal{C}(H)), \qquad b' \in B_p(\mathcal{C}(H')),$$

one has $b' \succsim_p b$. Thus the core satisfies RI-best for unbounded Shapley–Scarf housing markets.

Proof. Choose $X \in \mathcal{C}(H)$ with $X(p) = b$. By Theorem 3, there exists $Y \in \mathcal{C}(H')$ such that $Y(p) \succsim_p b$. Since b' is best for p among all objects attainable in new core allocations, $b' \succsim_p Y(p)$. Transitivity gives $b' \succsim_p b$. ☐

5 Conclusion

The corrected proof obtains a complete solution of the mathematical problem: the best object agent p can receive in a core allocation cannot become worse when p's endowed object is improved in the preferences of other agents. The proof is not independent new progress relative to the current literature, because the stronger theorem is already known. Its value is as a cleaned-up verification and repair of the draft argument: the dummy vertices are handled in a precisely defined auxiliary digraph, and the final claim excluding new cycles after dummy deletion is proved explicitly.

References

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